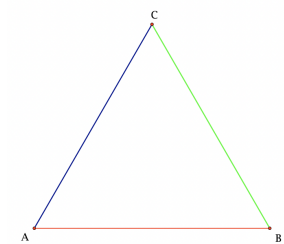


An Activity: What are the symmetries of an equilateral triangle?



- 1. Symmetry: a rigid motion of the plane (i.e., a motion which preserves distances) which also maps the triangle to itself. What are the symmetries you can find? How can we describe them?

- 2. How many symmetries did you find? Are there any more?? Why or why not??

- 3. What happens if we combine two symmetries?
 - $R_{120} \circ F_c$

 - $F_c \circ R_{120}$

$\circ$	R_0	R_{120}	R_{240}	F_A	F_B	F_C
R_0						
R_{120}						
R_{240}						
F_A						
F_B						
F_C						

4. What properties do you observe?

•

•

•

•

The Dihedral Groups of Order $2n$

Determine the symmetry group of each of the designs below.

